


Optimization problems

Objective: To arrange school Trip for students (400)

Transport company: 9 10 buses - Capacity of 50, cost: 800\$
8 buses - Capacity of 40; cost: 600\$

mathematical

Available drivers are 9.

min. the total cost

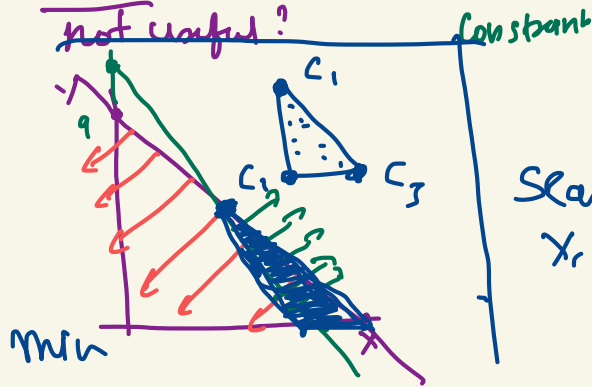
Linear Progra

$800x + 600y$.
 $x=0, y=0$ min the cost at 0

min $800x + 600y$

← cost of constraints

min
 $c_1, \dots, c_2$
 $c(c_1)$
 $c(c_2)$



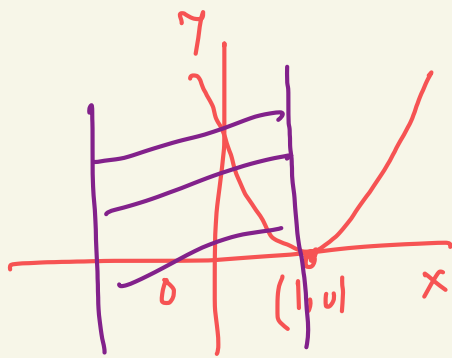
Search for x, y

$x + y \leq 9$
 $50x + 40y \geq 400$

How to solve this
 Graphically
 $x \geq 0, y \geq 0$

392
 406

$0 \leq x \leq 10$
 $0 \leq y \leq 8$



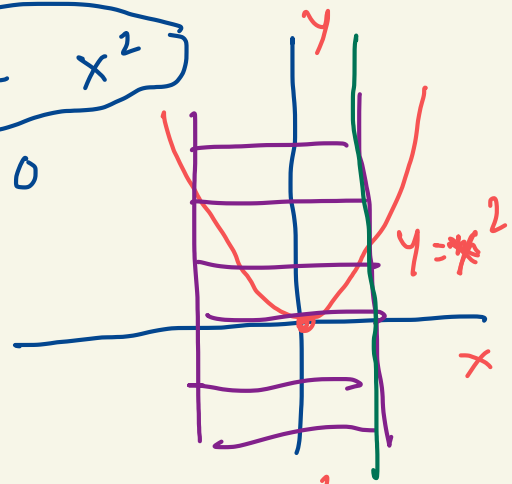
$$y = (x-1)^2$$

min value = 0
it occurs at $x=1$.

$-1 \leq x \leq 1$
doesn't change

$$y = x^2$$

0, 0



min x^2

min = 0
occurs at $x=0$

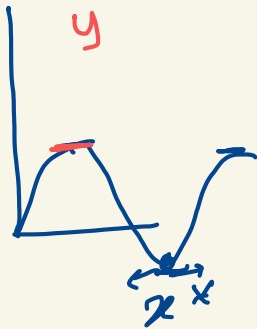
$$-1 \leq x \leq 1$$

$$x \geq 1$$

doesn't
change

local behavior of f along x^*
 x^* - local min.

$$\left. \frac{\partial f}{\partial x} \right|_{x=x^*} = 0$$



$$f(x^* + \alpha) = f(x^*) + \left. \frac{\partial f}{\partial x} \right|_{x=x^*} \cdot \alpha + \text{H.O.T.}$$

$$f(x^* + \alpha) - f(x^*) \approx \left. \frac{\partial f}{\partial x} \right|_{x=x^*} \cdot \alpha$$

$$\left. \frac{\partial f}{\partial x} \right|_{x=x^*} = -\alpha$$

$$f(x^* + \alpha) - f(x^*) = \underbrace{-\alpha^2}_{\text{negative.}}$$

$$f(x^* + \alpha) < f(x^*) \Rightarrow x^* \text{ is not a min.}$$

what
happens

$$\left. \frac{\partial f}{\partial x} \right|_{x=x^*} = 0$$

4.

$y = \sin x.$
 $\frac{\partial f}{\partial x} = \cos x.$
 $\frac{\partial^2 f}{\partial x^2} = -\sin x$

Second order condition.
 $\cos x = 0 \Rightarrow x = \pi/2, 3\pi/2$
 $\frac{\partial^2 f}{\partial x^2} = -\sin x$

which of these are min.
 is there a condition.

is there a condition.

$$f(x^* + \alpha) - f(x^*) = \cancel{f'(x^*)\alpha} + \frac{1}{2} f''(x) \cdot \alpha^2 + \dots$$

$$f(x^* + \alpha) - f(x^*) = \frac{1}{2} f''(x) \cdot \alpha^2 \Big|_{x=x^*}$$

sign of this ≥ 0

$f''(x) = \frac{\partial^2 f}{\partial x^2} \Big|_{x=x^*} > 0$

$$\frac{-2}{-3} < 0$$

$$\frac{2}{3} > 0$$

$$\lambda = -\frac{2}{3}$$

$$f(x, y) = (x-1)^2 + (y-2)^2$$

$$\begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \\ \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

first order

$$\begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix}$$

Hessian

H

$$\nabla f = 0$$

$$\begin{pmatrix} 2(x-1) \\ 2(y-2) \end{pmatrix} = 0$$

$\neq 0$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

> 0

positive def

Jacobian matrix

$x=1, y=2$
as the extremum points

$\Rightarrow$ minima.
eigen values are > 0 .

$$\begin{bmatrix} -2 & 0 \\ 0 & 2 \end{bmatrix}$$

> 0

< 0

No

No

$\text{tr def } M > 0$ eig values $\lambda_i > 0$ $\left[\begin{array}{cc} 3 & 0 \\ 0 & 0 \end{array} \right]$
 $M \geq 0 \rightarrow$ $\lambda_i \geq 0$
 tr semi definite
 when is a matrix > 0 (tr definite Matrix)

$H > 0 \Rightarrow \text{min.}$

$M < 0$ -ve definite

$M \leq 0$ -ve semi definite

$$\Delta f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} = 0$$

$$f(x_1, \dots, x_n)$$

$$\Delta^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

$$f(x_1, x_2) = x_1^2 \cos x_2^2 + x_2^L \sin x_1^2 + \tan \frac{x_2}{x_1}$$

$$\min f(x_1, x_2) \quad \underline{\hspace{15cm}}$$

Numerical Technique

Unconstrained optimization: Gradient descent
for Least Squares (linear regression)
 $(0, 6), (1, 0), (2, 0)$

$$y = mx + c$$

suppose: $m = -3$,

$$c = 7$$

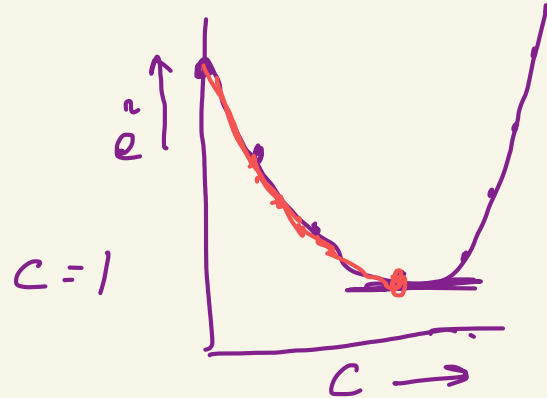
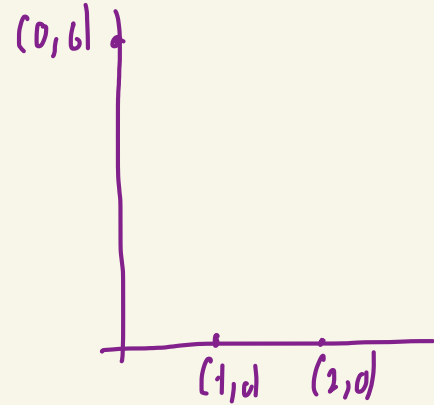
$$\alpha = 0.1$$

	Observed	Predicted	error
$(0, 6)$	6	0	36
$(1, 0)$	0	-3	9
$(2, 0)$	0	-6	36
			81
	6	1	25
	0	-2	4
	0	-5	25
			54

$$y = -3x + c$$

$$6 = -3 \cdot 0 + 0$$

function of 1 variable



$$e = (6-c)^2 + (3-c)^2 + (6-c)^2$$

Start with an initial guess

$$x(t) = x(t-1) - \alpha \nabla f(t-1)$$

↓ learning rate

initial guess let $c = 0$

$$\alpha = 0.1$$

$$c[1] = 0 - 0.1(-30) \Rightarrow c[1] = 3$$

$$c[2] = 3 - 0.1(-12) \Rightarrow c[2] = 4.2$$

$$c[3] = 4.2 - 0.1 \nabla f|_{c=4}$$

$$\downarrow$$

$$4.87 \dots \rightarrow 5$$

$$\nabla f =$$

$$-2(6-c)$$

$$-2(3-c)$$

$$-2(6-c)$$

$$\nabla f \text{ at } c=0$$

$$\nabla f|_{c=3}$$

m is unknown c is also unknown

$(0, 6), (1, 0), (2, 0)$

$[-3, 5]$

$$f = (6-c)^2 + (m+c)^2 + (2m+c)^2$$

$(0, 6)$	6	$m \cdot x + c$	$e_0 = [6 - c]^2$
$(1, 0)$	0	$m \cdot x + c$	$e_1 = [0 - m - c]^2 = (m+c)^2$
$(2, 0)$	0	$m \cdot x + c$	$e_2 = [0 - 2m - c]^2 = (2m+c)^2$
$[-2, 4] \quad d = 0.1$ $x(1) = \begin{bmatrix} -2 \\ 4 \end{bmatrix} - 0.1$			$\begin{bmatrix} 2(m+c) + 4m + c \\ -2(6-c) + 2(m+c) + 2(2m+c) \end{bmatrix} \quad (-2, 4)$

$$\begin{pmatrix} -2 & 4 \\ -2.4 & 4 \\ -2.4 & 4.2 \\ & \vdots \end{pmatrix}$$

$$-2.84$$

$$4.785$$

$$\begin{array}{c|c} -2.99 & \\ 4.98 & \end{array}$$

← after 10 iter

how to choose α

4
what is a
good initial
guess

4

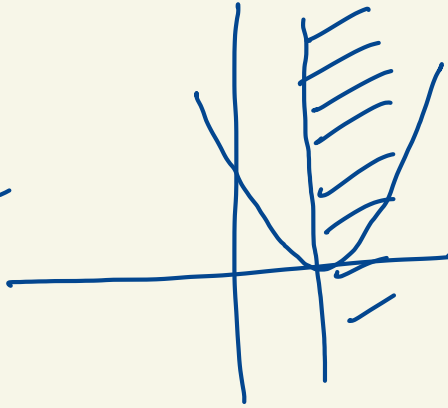
how many
iterations to run

S-G-D

$$y = x^2 \quad (x-1)^2$$

$$x \geq 1$$

Does not



min is same

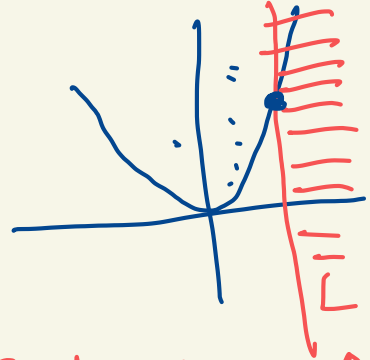
① without constraint

② with constraint

$$-1 \leq a \leq 1; \quad x \geq 1$$

$$y = x^2$$

$$x \geq 1$$



Earlier min = 0

does min change

min = y = 1
occurs at x = 1

min change $x \geq 1$.

$$\hat{x} = (A^T A)^{-1} A^T b$$

$$(A^T A) \hat{x} = A^T b$$